SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

C01_AT2 — DATA INTERPRETATION EXERCISE

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO1 –Apply (BL3)	**Assessment:**	Assessment Tool 2 – Data Interpretation
Weightage:	20% of CIA	**Total Marks:**	20(2 Questions × 20 Marks)
CO1 Statement:	*Understand the fundamental concepts, principles, and techniques of data visualization.(BL2)*		
Student Name:	RAGUL.M	**Reg. No.:**	1925240566
Section / Batch:	SLOT C	**Date:**	28.07.2026

Code of Conduct

I RAGUL.M/192524066 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: RAGUL.M

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.

Annexure A – DATA INTERPRETATION

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT2 — Data Interpretation Exercise

CO1 · BT3: Apply ·

This rubric is used to assess student responses to the AT2 Data Interpretation Exercise problems, in which students interpret a described data visualization (data types, aesthetic mappings, scales, coordinate systems, and color scales) and answer accompanying sub-questions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough, accurate understanding of data types (nominal, ordinal, numerical), aesthetic mappings, scales, coordinate systems, and color scales as they apply to the given visualization scenario.	Good understanding of the relevant visualization concepts, with only minor gaps in identifying data types, aesthetic mappings, or scale choices.	Basic understanding of visualization concepts, but with some misconceptions about data types, aesthetics, coordinate systems, or color scales.	Poor understanding of core visualization concepts; major misconceptions about data types, aesthetics, coordinate systems, or color scales.
Explanation	**25%**	Provides detailed, well-reasoned explanations for each sub-question, using specific details from the scenario and correct visualization terminology.	Provides clear explanations with adequate justification, referencing the scenario appropriately for most sub-questions.	Explanations are limited or superficial; lacks depth or fails to consistently connect reasoning to the given scenario.	Explanations are poor, unclear, or missing; answers are unsupported assertions with no reasoning shown.
Application	**20%**	Effectively applies visualization concepts to interpret the specific scenario, drawing accurate, well-reasoned conclusions from the described chart or data pattern.	Applies concepts to interpret the scenario correctly, with only minor errors in reasoning or in the conclusions drawn.	Limited application of concepts to the scenario; conclusions are weakly supported or only partially correct.	Fails to apply concepts to the scenario, or the conclusions drawn are incorrect or irrelevant to the data shown.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Organization	**15%**	Answers are well-structured, addressing each sub-question (a, b, c) in a clear, logical sequence with coherent flow throughout.	Answers are organized and address each sub-question, with only minor issues in flow or clarity.	Basic structure; sub-questions are addressed but answers lack clarity or a logical sequence.	Poorly organized; answers are incoherent, and sub-questions are left unaddressed or answered out of order.
Accuracy	**10%**	All parts of the answer — data type identification, aesthetic/scale justification, and final interpretation — are completely accurate.	Mostly accurate, with only minor omissions or a small error in one part of the answer.	Partially correct; some key points (e.g., data type, aesthetic justification, or interpretation) are missing or incorrect.	Answers are largely incorrect, with major gaps in identifying data types, aesthetics, scales, or drawing conclusions.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

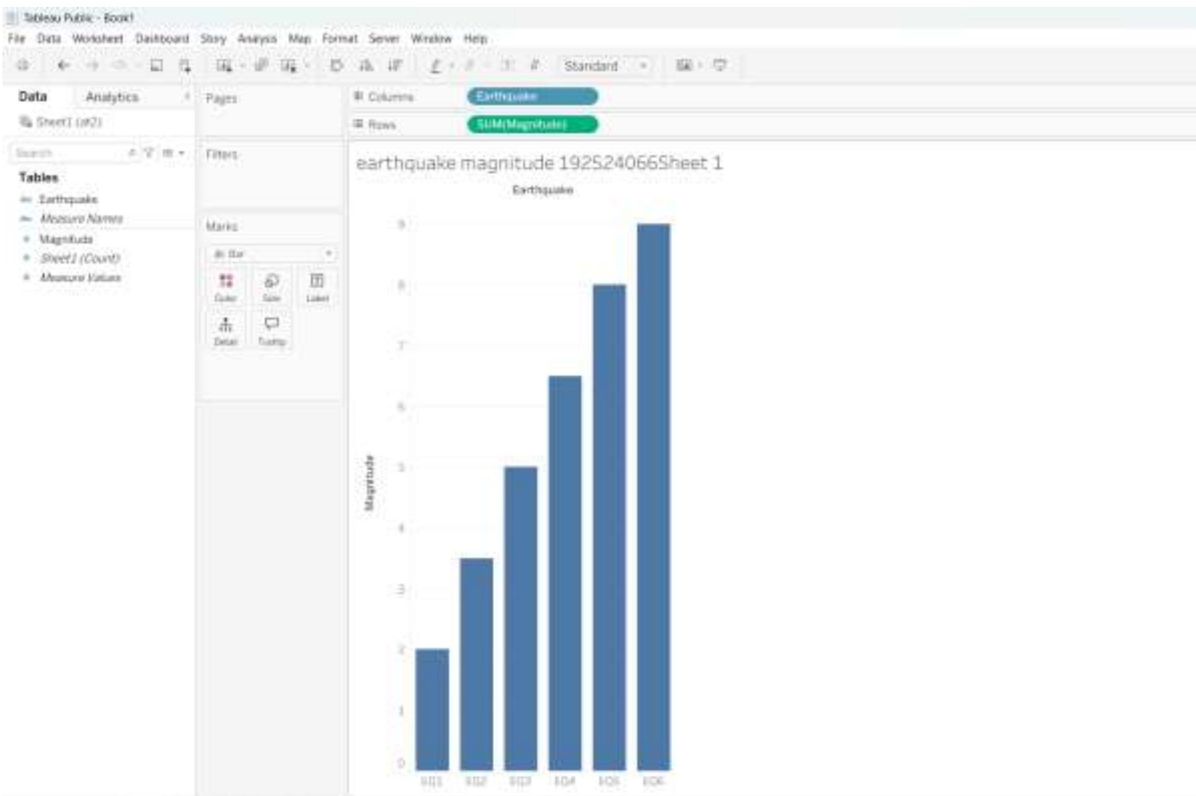
Problem 9: Earthquake Magnitude Bar Chart

Objective

To visualize earthquake magnitudes using a bar chart and understand how earthquake magnitudes are represented on a standard linear Y-axis.

Description

This visualization presents earthquake magnitudes ranging from 2.0 to 9.0 on the Richter scale. The X-axis represents different earthquakes, while the Y-axis represents their magnitude values. A bar chart is used because it allows easy comparison between the magnitudes of different earthquakes. The height of each bar corresponds to its magnitude value.



Observation

* Earthquake magnitudes increase from 2.0 to 9.0.
* Taller bars indicate stronger earthquakes.
* The chart uses a linear Y-axis, so the difference between bars appears equal.
* The visualization is simple and easy to compare, but it does not fully represent the real-world increase in earthquake energy because the Richter scale is logarithmic.

Conclusion

The bar chart effectively compares earthquake magnitudes visually. It helps identify the strongest and weakest earthquakes quickly. However, since the Richter scale is logarithmic, a small increase in magnitude actually represents a much larger increase in earthquake intensity than the chart visually suggests.

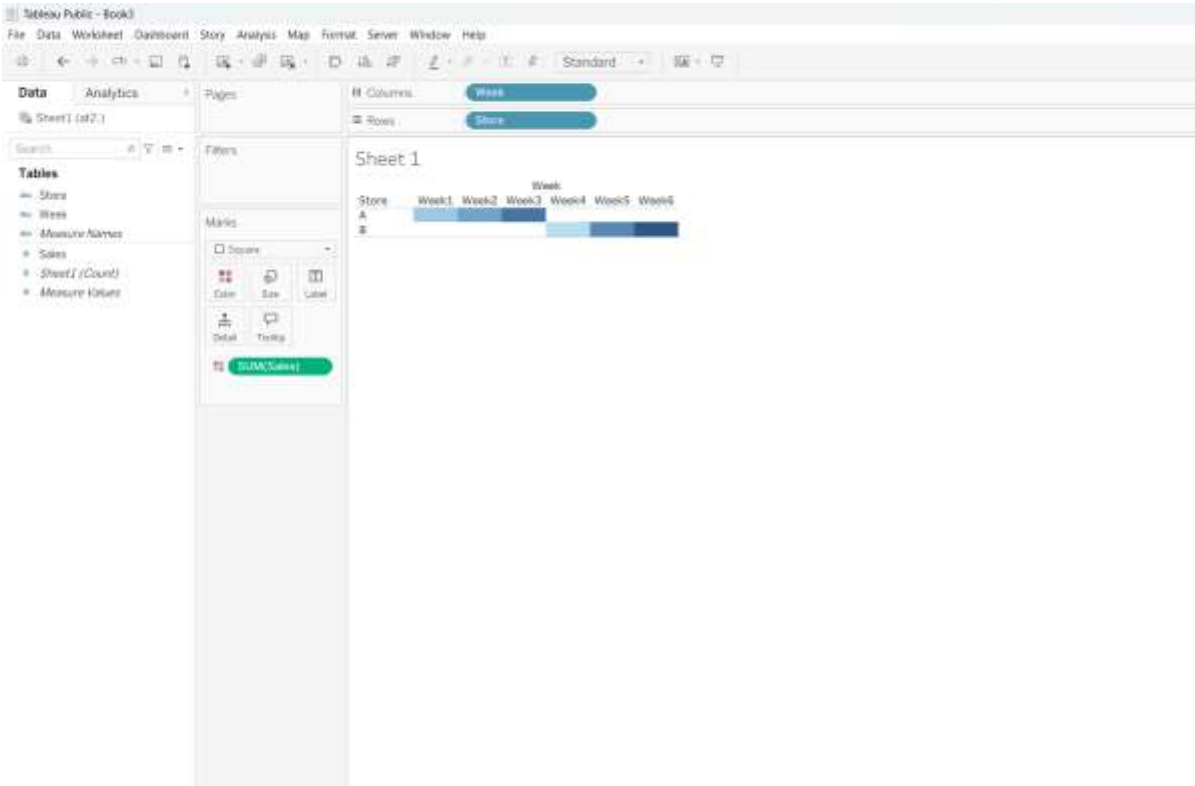
Problem 19: Weekly Sales Heatmap

Objective

To visualize weekly sales data of different stores using a heatmap with a sequential blue color scale.

Description

The heatmap displays weekly sales for multiple stores. The X-axis shows the weeks, and the Y-axis shows the stores. Sales values are represented using different shades of blue. Light blue indicates lower sales, while dark blue indicates higher sales. This visualization helps users identify sales trends without reading every numerical value.



Observation

* Dark blue cells represent the highest sales.
* Light blue cells represent the lowest sales.
* Sales performance can be compared easily across different stores and weeks.
* A darker cell immediately attracts attention and indicates better sales performance.

Conclusion

The heatmap is an effective way to compare weekly sales across stores. The sequential color scale makes it easy to identify high and low sales at a glance. This visualization improves data interpretation and helps users quickly recognize important sales patterns and trends.